

Source → Codecademy and GPT

1 HTML → Hyper Text Markup Language

↳ Means that text that can be connected to other doc / pages using links

Ex - `<a href="https://google.com">G</a>`

2 Markup Language → Means Using Tags to tell the browser what the content is and how it should be understood.

`<h1> My Website </h1>`

`<p> I am Aman </p>`

3 Semantic → Means Meaning or Purpose

↳ A semantic element tells the browser and the developer what the content is meant for.

Ex → `<h1> My Portfolio </h1>`

↳ Tells "This is the Main heading of the page".

4 Tag → A tag is a Markup Used to tell the browser what type of content an element contains

Ex - `<p> Hello </p>`

↳ Opening Tag

↳ Closing Tag

5 Element → An HTML element is the complete structure that consists of the tag(s) and the content inside them

Ex - `<p> I am Aman </p>` → The entire thing is an element

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Heading Tags

→ 6 heading Tags

<h1>	Main heading	</h1>
<h2>	Main Subheading	</h2>
<h3>	Subsection Under <h2>	</h3>
<h4>	" " "	</h4>
<h5>	" " "	</h5>
<h6>	Least imp heading level	</h6>

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Paragraph Tag → Define Paragraph OF text in HTML

ex - <P> My name is Aman </P>

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 → Line break { Move the following Content to a new line. }

ex - <P> My name is Aman.

I am learning HTML.

I want to become a Web developer

</P>

Output →

My name is Aman.

I am learning HTML

I want to become a Web developer

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<hr> →

Horizontal Rule

→ It separates the two Section by using horizontal line.

ex- `<h1> About me </h1>`
`<p> I am learning Web dev . </p>`
`<h2>`
`<h2> My Skills </h2>`
`<p> HTML, CSS, JS </p>`

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Unordered list - `<ul>` Used When Order doesn't Matter

ex- `<ul>`

`<li> Aman </li>`
`<li> Ankit </li>`
`<li> RaJu </li>`
`</ul>`

Output → ~~•~~ • Aman
• Ankit
• RaJu

`<li>` Means list

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Ordered List → `<ol>` Used When Order Sequence Matters

Output

ex → `<ol>`

`<li> Aman </li>`
`<li> Ankit </li>`
`<li> RaJu </li>`

`</ol>`

Output → 1. Aman
2. Ankit
3. RaJu

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Anchor Tag → Used to Create Hyperlinks `<a>` `</a>`
↳ Redirect to another Page.

ex - ` Go to google
`

href attribute → Hypertext Reference it tells the browser where the link should go.

ex - ` Youtube
`

Other Useful `<a>` attributes

(i) target → Controls where the link opens

ex → `<a href = "https://www.google.com" target = "_blank"> Google </a>`

target = "_blank" → Opens the link in new tab.

(ii) title → Provides additional information, usually shown as tooltip when you hover over the link.

ex → ` Google
`

(iii) Download → For downloading the document

ex → `
Download Resume `

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Image Tag `<img>` → Used to display an image on a webpage

ex - `<img src = " Photo.Jpg" alt = " my Photo" >`

Important attributes

(i), `src` - Source → tells the browser where the image is located.

ex - `<img src = " Photo.Jpg" >`
We can also use image URL:

(ii), `alt` → Alternative Text → Provides text description of the image.

ex - `<img src = " Photo.Jpg" alt = " Aman's Profile Pic" >`

It is Useful When → The image cannot load
Screen readers need to describe the image

(iii), `Width and Height`

ex - `<img src = " Photo.jpg" alt = " my Photo" width = " 300" height = " 200" >`

Note → Attributes provide an additional information or properties to an HTML element.

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Nested list → Means a list inside another list

ex-

 Frontend

 HTML

 CSS

Output →

- Frontend

- HTML

- CSS

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Table → Used to display data in rows and columns

ex →

Name	Age	Course
Aman	24	MCA
Akhil	20	Science

Structure →

<table border = "1" >

<tr> → Table row

Table header <th> Name </th>

<th> Age </th>

<th> Course </th>

</tr>

<tr>

<td> Aman </td>

<td> 24 </td>

//_

```
<td> MCA </td>
</tr>
```

Same for ankit

```
</table>
```

Used Semantic tags in Table

ex

Item	Quantity	Price
Apple	10	20 ₹
Orange	5	30 ₹
Total	15	60 ₹

For this Use `<thead>` `<tbody>`

And `<tfoot>`

Footer Tag.

ColSpan $\Rightarrow$ Allows a Cell to Span Multiple Column

ex $\Rightarrow$ `<th colspan="2"> Student Details`
`</th>`

(Cell occupies 2 Columns)

Rowspan $\Rightarrow$ Allows a Cell to Span Multiple Rows.

ex $\Rightarrow$ `<th rowspan="2"> Arman </th>`

16 File Path

(1) Same Folder - Relative Path

Suppose → Project /

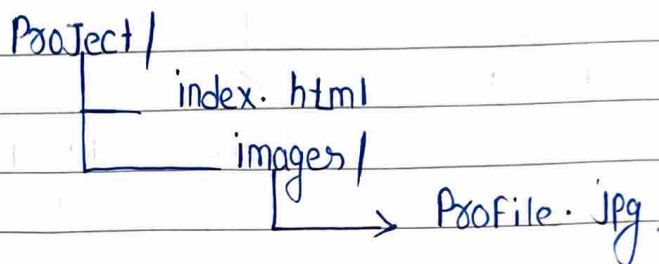


in index.html:

ex- `<img src = "Profile.jpg" alt = "Profile">`

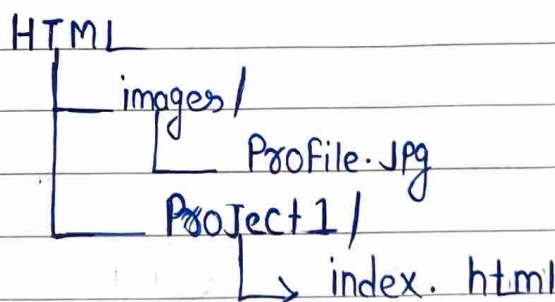
Because both files are in the same folder

(ii) File inside a folder



ex- `<img src = "images/Profile.jpg" alt = "Profile">`

(iii) Go one folder back - ../



From Project 1 / index.html:

`<img src = "../images/Profile.jpg" alt = "Profile">`

.. means go one folder up.

(iv) Linking Another HTML File

```
Project /  
├── index.html  
└── about.html
```

ex. `<a href = "about.html" > About me </a>`

Note Avoid Using your Computer Absolute Path like

```
<img src = " / Users / aman / DeskTop / HTML /  
Profile . jpg >
```

This will generally work on your own computer
And it is not suitable for a website / Github Project

Absolute Path

→ Complete Path from the root location to the File

Note → Used `./` = Current folder

```
Project /  
├── index.html  
└── images /  
    └── Photo . jpg
```

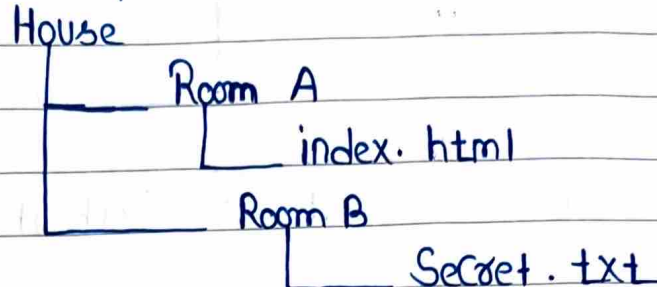
From index.html

```
<img src = ". / images / Photo . jpg" alt = "Profile">  
<img src = "images / Photo . jpg">
```

Both mean - Start from the Current folder

Why browser Cannot Access Absolute Path

Imagine your Computer is a house



Your browser is saying index.html Now Suppose the HTML Says

`<img src = "/Users/amanraj/Secret.txt">`

The browser Says :

❌ Wait ! A Website Should not be allowed to Open any Random File on your Computer

Because if browser allowed Websites to access any file a Website Could try

`/Users/amanraj/Documents/bank.pdf`

Password or any Sensitive info

So the browser Cannot Freely Access arbitrary files on your Computer.

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HTML Boilexplate { HTML 5 }

Why Use Boilexplate ?

→ Ready made basic Structure Of an HTML document

→ (i), Tells the browsers to Use modern HTML

`<!DOCTYPE html>`

(ii), Provides the Correct document Structure

html

`<head>`

`<body>`

(iii), Helps text display Correctly

`<meta Charset = "UTF-8">`

(iv), Makes the Page Work properly on mobile devices.

`<meta name = "Viewport" Content = "Width = device-Width, initial-Scale = 1.0">`

(v), Gives your Page a browser-tab title

`<title> my Website </title>`

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Div

`<div>` means division

Generic Containers Used to group related HTML elements together.

ex. <div>

<h2> About me </h2>

<p> I am aman </p>

</div>

Think like this

About me
I am learning web-dev

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Class

↳ An attribute used to give an element a name/group

ex. <div class = "Card">

<h2> My Profile </h2>

<p> Developer </p>

</div>

Here class name Card Later CSS can target
• Card:

CSS

• Card { background: gray; }

Note - Multiple elements can have the same class

<div class = "Card"> Profile </div>

<div class = "Card"> Skills </div>

Class

Think ~~Class~~ = group / category

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id
 ↳ Gives an element a Unique identifier
`<div id = "Profile" >`
`<h2> Aman </h2>`
`</div>`

Think ID = Unique identity

ex - `<div id = "name" ></div>`
`<div id = "Skills" ></div>`

In CSS Selector We Use # Profile for id.

Can We Use both ?

yes!

`<div id = "Profile" Class = "Card" >`
`<h1> Aman </h1>`
`<p> Java Developer </p>`
`</div>`

Here id = "Profile" → Unique identity
 Class = "Card" → belongs to the Card group

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`<header>`
 ↳ Intro/ top information

ex - Logo, Website name, main heading,
 Navigation

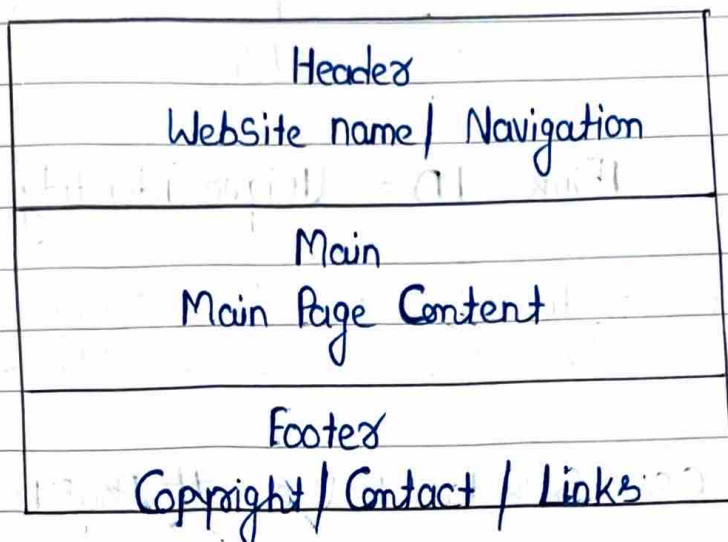
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`<main>` Contains the actual main Content
 A Page normally has one `<main>` element

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Footer < Footer >

→ bottom/end info of a web-page
or Section



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HTML Forms

→ Used to Collect info from the
User

Structure

< form >

< label for = "name" > Name
< /label >

< input type = "text" id = "name" name =
"name" >

< /form >

(i) < label > → Provides a description for a form
Control.

The for attribute connects
the label to an input.

//_

ex - `<label for = "name"> First Name </label>`
`<input type = "text" id = "name">`

If you Click First Name, the browser Can focus the input.

(ii), `<input>` - It Collect different type of User info

(a) Text input $\rightarrow$ `<input type = "text">`

(b) placeholder $\rightarrow$ Gives the User hint/example about what to enter.



(iii), Value $\rightarrow$ Represent the actual Value
`<input type = "text" Value = "Aman">`

(iv), id $\rightarrow$ Temp for Connecting the level

(v), name $\rightarrow$ Think like name Of the data.

`<input type = "text" name = "Username">`

Suppose the User enters: Aman The Submitted data becomes Username = Aman.

(vi), Required $\rightarrow$ Tells the browser this is mandatory
You must Provide Value.

`<input type = "email" Required>`

(vii) Email input → `<input type = "email" >`

(viii) Tel input → `<input type = "tel" >`

(ix) Date input → `<input type = "date" >`

(x) Password input → `<input type = "password" >`

Browsers hides the Characters while typing

Imp → type "Password" does not encrypt the Password.

(xi) Radio buttons → Allows Users to Choose one Option from a group

ex - `<input type = "radio" id = "male" name = "gender" Value = "male" >`

`<label for = "male" > male </label >`

Same for Female and Others

Rule → All radio button in the same group have the same name.

like name = "gender"

Male Female Others

Value in radio buttons → Value represents what should be submitted if that option is selected.

(xii) Checkbox → Allows Users to Select Zero, one or multiple Options.

//_

Ex - `<input type = "checkbox" id = "Sports" name = "interest" value = "Sports" >`
`label for = "Sports" > Sports </label >`

Also We Can add more Options

(Xiii, File Upload $\rightarrow$ Allow Users to Upload a File
`<input type = "file" name = "Photo" accept = "image/*" >`

$\rightarrow$ Tells that Field is intended for image Files.

(Xiv, `<textarea >` $\rightarrow$ Used for multiline text.

`<textarea`

`id = "address"`

`name = "address"`

`rows = "4"`

`cols = "40" > </textarea >`

(Xv, `<Select >` $\rightarrow$ Creates a dropdown menu.

`<Select name = "Course" >`

`<option value = "Cse" > Cse </option >`

`" " "it" > it < "`

`" " "Me" > "me" < "`

`</Select >`

Placeholder option in Select

`<option value = " > -- Select -- <`

`</option`

(Xvi) Submit button $\rightarrow$ `<button type = "Submit" >`
Registered `</button >`

(Xvii) Reset button $\rightarrow$ `<button type = "reset" >` Clear
`</button >`

(Xix) action $\rightarrow$ `<form action = "/registered" >`

Action tell the browser where should be the form data sent

(XX) method $\rightarrow$ Tells the browser how to send the form data.

Two imp methods are
Get and Post

Ex $\rightarrow$ ~~`<form method = "get" >`~~
`<form action = "/registered" method = "post" >`

Get Vs Post

Get $\rightarrow$ `<form method = "get" >`

Data is generally included in the URL
like `/search ? name = Aman & Course = CSE`

Post $\rightarrow$ Data is sent in the HTTP request body.
like - `FirstName = Aman`
`email = aman@example.com`

(Xxi)

Semantic Grouping $\rightarrow$ `<FieldSet>`
When forms become large, we can group related controls.

`<Fieldset>`

`<legend>` Personal Details `</legend>`

`<label for = "name">` Name `</label>`

`<input type = "text" id = "name">`

`</Fieldset>`

`<legend>` gives that group a title

HTML Form Validation

$\rightarrow$ Checking whether the data entered by the user is acceptable before the form is submitted.

(i) Required $\rightarrow$ Users must provide a value

`<input type = "text" name = "firstName"`

`Required >`

(ii) minlength $\rightarrow$ Specifies the min no of characters allowed

`<input type = "text" name = "Username"`

`minlength = "3" >`

like example with pass

(iii) maxlength $\rightarrow$ Max no of characters allowed.

ex `maxlength = "20"`

Note $\rightarrow$ min and max specifies the numeric/date value

Example $\rightarrow$ `<input type = "number" name = "age"`
`min = "18" >`
Same for max

Also Used in date Value

`<input type = "date"`
`name = "dob"`
`min = "2000-01-01"`
`max = "2010-12-31"`
`>`

(iv), ~~(v)~~ Pattern $\rightarrow$ `<input`
`type = "text"`
`name = "Username"`
`Pattern = "[A-Z a-z]+" >`

This means Only letters from A-Z or a-z

Another Example $\rightarrow$ Indian Style 10 Digit Phone
`<input`

`type = "tel"`
`name = "phone"`
`Pattern = "[0-9]{10}" >`

Means `[0-9]` $\rightarrow$ any digit from 0 to 9
`{10}` $\rightarrow$ exactly 10 times

~~(v)~~, (v) readonly $\rightarrow$ The User Can see the Value
but Cannot modify it

ex - `<input type = "text"`
Value = "Aman"
readonly >

(vi) disabled → The Users Cannot interact With this Form Control.

(vii) autoComplete → Suggest or automatically Field previous Entered info.

(ix) Checked → Used with radio button and Checkboxes
It makes the Control Selected by default.

ex → `<input`
type = "radio"
name = "gender"
Value = "male"
checked >

`<label>male </label>`

When the page loads -
● male already Selected
○ Female
○ Other

(x) Selected → makes an `<Option>` default Checked Option

ex - `<Select Name = "Course" >`

`<Option Value = "CSE" >CSE </option>`

`" " "it" Selected >it <`
`/option>`

`</Select>`

(xi),

disabled On `<Option>` → We Can also disable an Option.

```
< Select Name = "Course" >
```

```
<Option Value = " " disabled Selected >
```

```
- - Select Course - - </Option>
```

```
<Option Value = "Cse" > Cse </Option>
```

```
</Select>
```

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HTML Media Tag

(i),

Audio tag → To add audio

```
<audio Controls Src = "Song.mp3" > </audio>
```

Control tell the browser

└ Show audio Controls to the User

Audio tag With Source

└ Multiple media formats

```
<audio Controls>
```

```
< Source Src = "Song.mp3" type = "audio/  
mp3" >
```

```
< Source Src = "Song.ogg" type = "audio/ogg" >
```

Your browser does not Support audio Playback.

```
</audio>
```

//_

Imp Point → You don't need to have multiple format for every audio file. GF you have Song.mp3
Just do

<Audio Controls>

<source Src = "Song.mp3" type
= "audio/mpeg">

(ii)

Video Tag → Basic Structure

<Video Controls Src = "Video.mp4">

You can also Specify dimensions :

<Video

Controls

Width = "600"

Src = "Video.mp4">

</Video>

Controls Used for Play/Pause Full screen etc

a

autoplay → <Video autoplay> Attempt to Start automatically

b

autoplay muted → <Video autoplay muted>
Start the Video without muted.

c

loop → <Video Controls loop> → When the Video finishes Video Start again.

d

poster → Specifies the image shown before the Video Starts like thumbnail

<Video

Controls

Poster = "thumbnail.jpg" Src = "Video.mp4">
</Video>

< Video > With < Source >

< Video Controls >

< Source Src = " Video.mp4" type = " Video/ mp4" >

< Source Src = " Video.Webm" type = " Video/Webm"
Your browser does not support Video Playback >
< /Video >

Because diff browsers / devices may support diff media formats The browser checks and uses a supported one.

e

Width and height → < Video

Src = " Video.mp4 "

Controls

Width = " 600 "

heights = " 400 "

< /Video >

< iframe > - inline frame

Used to embed another webpage to your website.

< iframe Src = " https:// example.com " > < /iframe >

Important < iframe attributes >

a

Width → Controls the Width

< iframe

Src = " https:// example.com "

Width = " 600 " >

< /iframe >

b height → Control the height

c title → Very imp for accessibility.

Suppose you have a youtube Video

```
<iframe
```

```
width = "560"
```

```
height = "315"
```

```
src = "https://www.youtube.com/embed/abc123"
```

```
title = "Youtube Video Player"
```

```
allowFullscreen >
```

```
</iframe>
```

Can we embed any website?

→ No bcoz some website prevent themselves from being displayed in iframe.

Multipage Website

Example → My Website

index.html	Home
about.html	About
services.html	Services
projects.html	Projects
contact.html	Contact

Use <Nav> box for putting Page links